Forensics for FBI Agents (Late 1990s)

Introduction

FBI field agents in a Violent Crime Squad are expected to possess a solid working knowledge of forensic science. While they are not laboratory scientists, agents know how to secure crime scenes, recognise and collect evidence, and understand the capabilities and limitations of forensic analyses. This sourcebook provides a detailed overview of forensic procedures and techniques as of 1997, focusing on those most relevant to violent crimes. Real terminology is explained in plain language so that you can both “talk the talk” like an FBI agent and understand how the underlying methods work. All information reflects historically accurate practices up to 1997 – any techniques developed later are omitted as they would not yet exist in our setting.

Topics Covered: We’ll cover general crime scene procedures and evidence handling, then delve into specific forensic disciplines: fingerprints, DNA, ballistics, trace evidence (hairs, fibres, etc.), blood pattern analysis, autopsy and pathology basics, forensic entomology (insect evidence), anthropology (skeletal remains), odontology (dental evidence), questioned documents, toxicology, and even a note on behavioural analysis support. Finally, we highlight exotic or region-specific forensic considerations that a Phoenix-based agent should know – for example, how the desert climate affects evidence and decomposition. By mastering these topics, you’ll be equipped to make realistic decisions at a crime scene and during investigations, just as an actual FBI agent in 1997 would.

Crime Scene Management and Evidence Handling

A violent crime scene is often chaotic, but the responding FBI agents must impose order and follow strict procedures to preserve evidence. Field agents are trained in Evidence Response Team (ERT) protocols, ensuring that vital clues are not lost. Below we outline how agents manage a crime scene from arrival to evidence transfer, emphasising techniques an agent in 1997 would use.

Securing the Scene and Initial Response

Upon arriving at a potential crime scene, an FBI agent’s first duty is to ensure safety and secure the area. This means checking that any suspects are no longer a threat, aiding any victims in need of urgent medical care, and then roping off the scene with tape. No unauthorised personnel – including curious onlookers or even fellow officers not involved in the investigation – should cross the perimeter. This prevents contamination of evidence. Agents will post guards or use police partners to maintain a security perimeter.

Inside the secured area, the primary rule is: touch nothing until it’s documented. Early-arriving agents do an initial walk-through to observe potential evidence (blood pools, shell casings, overturned furniture, etc.) without moving or disturbing anything. They mentally note transient evidence that could disappear if not promptly preserved – for example, a footprint in loose sand or a faint shoe print on a tile floor that might be smudged. Environmental conditions in Phoenix (like a gust of desert wind or the intense sun) might threaten certain evidence, so agents act quickly if needed. For instance, if there are footprints in sand outside, agents might photograph and cover them (with a box or tarp) to shield from wind until they can be cast. If bloodstains are exposed to the brutal sun, agents know to collect or shade them promptly, since heat and UV light can degrade DNA rapidly.

Agents also coordinate with local law enforcement at this stage. In many cases, Phoenix Police or county deputies might be first on scene; an FBI violent crimes agent will assert federal authority if it's an FBI case (e.g. federal kidnapping, a crime on federal land, or a serial killer case crossing state lines) but will work with local officers. They'll ensure that whoever arrived first has maintained the scene integrity. If not, the FBI agent may have to contend with inadvertent contamination (like a well-meaning cop who moved a weapon or covered a body). They'll document any such disturbances in their notes.

Documentation: Photography, Sketches, and Notes

Before evidence is collected or bodies moved, everything must be documented thoroughly. Typically, one of the team – often a trained evidence photographer – will take photographs of the scene. In 1997, this means using a 35mm film camera (digital cameras were not yet in use) with a flash for indoor or low-light shots. Standard practice is to capture wide-angle overviews of the scene (showing how the body was found in relation to the room, for example), then mid-range shots, and close-ups of specific evidence. Each piece of evidence (a shell casing, a knife, a shoeprint, etc.) is photographed in place, often both with and without a measuring scale in the frame for reference. Agents carry small rulers or place cards for this purpose. If needed, oblique lighting (flashlight at an angle) is used to highlight impressions or trace evidence.

Alongside photographs, agents will sketch the scene. A crime scene sketch is a rough diagram (later turned into a scale drawing) showing the layout and key evidence positions. It includes measurements (using a tape measure) from fixed points (walls, corners) to each piece of evidence, so that the exact location is recorded. This is crucial if something gets moved or if the scene needs to be reconstructed later. The sketch also notes the orientation (a North arrow) and other details like the positions of doors, windows, furniture, etc. In a desert outdoor scene, a sketch might use landmarks (a particular cactus, a rock formation) or stakes driven into the ground as reference points.

Agents take extensive notes throughout – either in a pocket notebook or on evidence log forms. They record the date, time, and conditions (weather can be important – e.g. "112°F and sunny, dry, slight wind from west" for a Phoenix summer crime scene). They note who is present and any initial observations (like "Victim found face-down on living room carpet, large pool of blood under head, apparent gunshot wound to back"). Every action gets logged: "10:35 AM – Det. Smith commenced photography. 10:50 – Body pronounced dead by coroner's investigator, ME's office notified." These contemporaneous notes help establish the timeline and are invaluable later for reports or court testimony.

Evidence Collection Techniques

In general, field agents at a crime scene will secure the scene, protect it, and seize obvious items during arrests / searches (guns, drugs, documents) – but they do not normally do the meticulous forensic processing. Instead, by the mid-1990s the FBI has formalised Evidence Response Teams (ERTs) to give each field division a dedicated crime-scene processing capability. ERTs are multi-disciplinary teams (mostly special agents plus professional, non-agent staff) trained to document scenes and recover / package evidence for the FBI laboratory. Members hold ERT as a collateral duty – you're still a field agent, but you get called when your team deploys to a crime scene. On joint scenes, it's common for the local Crime Scene Unit (CSU, e.g. Phoenix PD crime lab) to run primary processing under its Standard Operating Procedure (SOP), with FBI ERT augmenting (extra personnel, specialised mapping, alternative-light, sifting, excavation, etc.). On federal-lead matters (bombings, kidnappings, serial murders, reservation major crimes, terrorism), FBI ERT may run the scene, with locals assisting. Either way, it's the specialists, not the field agents, doing the fine work (unless the field agents are part of the ERT, obviously). So, for the sections below, when a crime scene technique mentions 'agents', it refers to the specialists (ERT, or CSU, although local PD's SOP may be slightly different and possibly slightly less stringent compared to the FBI's ERT).

Once documented, agents carefully collect physical evidence. Each item is handled with clean gloves (changing gloves between collecting different items to avoid cross-contamination). Standard FBI practice is to use appropriate tools: tweezers for small trace items, latex gloves for larger items, and sterile swabs for fluids. Evidence is placed in separate containers – usually paper bags or envelopes for most items, as paper is breathable (important for wet or biological stains so they don't mould). Plastic bags are generally avoided for anything with moisture (blood, for example) because sealing wet evidence can degrade it. A wet blood-stained cloth might be air-dried first or stored in a breathable paper package. Hard, dry items (a spent cartridge casing, a knife) can go in plastic if dry, but often paper is still preferred to prevent condensation. Each container is labelled with the date, time, case number, item description, and collector's name or initials.

So, what do field agents do on a crime scene?

First in: make safe, render aid, freeze the scene, log entries, and call ERT/CSU

Case direction: the case agent (the lead investigator in a criminal case, likely one of the field agents whose case the crime scene relates to) coordinates with the ERT/CSU lead (who is the scene manager for processing)

Seizures during ops: during a search warrant or arrest, agents do seize items listed / obvious contraband (guns, ledgers, cash, drugs, etc.) and maintain chain-of-custody – but they don't stop to dust a room or micro-collect fibres; they'll hold the location for ERT if forensic processing is needed.

Documentation & custody: agents photograph quick "as-found" overviews if needed for safety / record, then defer to ERT for full photography / measurements; agents complete FD-192/302 reports and evidence inventories, ERT handles forensic packaging and seals for the lab.

Evidence types encountered in violent crimes include weapons (guns, knives, blunt objects), biological samples (blood, saliva, semen, hair), latent fingerprints, bullet casings or shotgun shells, bullet fragments, fibres from clothing or furniture, documents (notes, IDs), electronic devices (in 1997 possibly pagers, answering machine tapes, maybe floppy disks), among others. Agents must know special collection methods for certain evidence:

- **Latent fingerprints:** These are invisible prints left by skin oils. At the scene, agents or techs might dust non-porous surfaces (glass, glossy furniture, vehicles) with fine powder and use tape to lift prints. Alternatively, they might use on-site cyanoacrylate fuming (superglue fuming) for hard surfaces: heating superglue in a chamber or bag with the object causes fumes that adhere to fingerprint residues, turning them white and visible. For porous materials like paper or drywall, chemicals like ninhydrin (which reacts with amino acids in sweat, turning fingerprints purple) might be applied – though typically that is done in a lab setting, not at the scene. If an item is small enough (a paper, a bottle, etc.), often agents will just carefully wrap and send the whole item to the FBI Lab for latent print development, rather than attempt it in the field. Fingerprints on human skin, such as on a strangulation victim's neck, are notoriously difficult to develop (skin is porous and prints don't last long). Research in the early '90s looked at special superglue fuming techniques for skin, but even by 1997 there's no reliable field method – agents mostly rely on other evidence in such cases.

- **Biological fluids (DNA evidence):** Blood, semen, saliva, or other body fluids are collected using sterile cotton swabs or by taking the whole item bearing the stain (cutting out a piece of bloodstained carpet, for example). Each stain is collected separately to avoid mixing. Swabs are air-dried if possible and then packaged in paper envelopes. Agents also make sure to control for contamination – using clean disposable instruments, and collecting a "substrate control" swab (an unused swab of a nearby unstained area) for lab comparison, in case the background material could interfere with tests.

- **Firearms and ammunition:** If a firearm is found, agents pick it up carefully by the grip or edges (to avoid smudging prints). First, they make it safe (unload it) unless it must be left loaded for some reason. They note the position of safety, hammer, etc., as found. Each gun is tagged, and the magazine and any chambered cartridge are also collected. Bullets and casings on scene are picked up with gloved fingers or tweezers (making sure not to scratch any markings). A neat trick: if bullets are lodged in walls or wood, agents or crime scene techs will remove the section of material around it rather than prying the bullet out with metal tools which could damage the striations.

- **Trace evidence:** Tiny bits like hairs, fibres, glass shards, paint chips, or gunshot residue are collected with great care. Hairs and fibres might be picked up with clean tweezers or using a tape lift or specialised vacuum with filter. Each trace evidence type is stored in a small druggist's fold paper or a vial. For clothing, often the entire garment is seized rather than picking off every fibre at the scene. Agents handling a murder victim's body might carefully wrap the hands in paper bags before transport to preserve any evidence under the nails or on the hands (like hair or fibres from the assailant). Footprints or tire tracks, if present in soil, are photographed and then cast using dental stone or plaster – a mixture is poured into the impression, allowed to set, capturing the 3D tread pattern.

Throughout collection, *chain of custody* is scrupulously maintained. This means every transfer of evidence is documented so it can be accounted for from scene to courtroom. As soon as an item is bagged, an evidence label is filled out and attached, and the agent writes their initials half on tape/half on bag to seal it (any tampering would be visible). A log sheet lists each item and who collected it. If evidence is handed off (say from an FBI agent to the Phoenix Police Department lab courier or to the FBI Laboratory), there's a receipt or record of that handover. Maintaining chain of custody is critical; if there's any break or unexplained gap, the defence could challenge the evidence's integrity in court.

Field vs Lab: An FBI field agent's job is mostly to recognise and preserve evidence; the detailed analysing is usually done back at the lab by forensic scientists. The FBI Laboratory (in 1997, located at FBI Headquarters in Washington, D.C.) provides a wide range of forensic services to support investigations. Agents in Phoenix would routinely send evidence to the FBI Lab or occasionally to the city's own forensic lab for processing. The FBI Lab has specialised units for DNA, fingerprints, firearms, tool marks, trace evidence, questioned documents, and more – and its services are available free to state and local agencies as well. As a field agent, you'd know generally what the lab can do and how to ask for the right analysis (for example, requesting fibre comparison on a piece of carpet fibre found on a victim, or asking the lab to match a recovered bullet to a suspect's gun). You'd also be aware of the turnaround time – some analyses (like a straightforward fingerprint search) might come back in days, whereas 1990s DNA testing could take weeks.

Working with the Medical Examiner: In violent crimes involving death, the local Medical Examiner (M.E.) or coroner plays a key role at the scene regarding the body. FBI agents do not move or examine the body internally – that's the M.E.'s job. Instead, agents will observe and sometimes photograph the body in place, and make note of things like lividity (blood pooling in body, which can suggest if a body was moved postmortem), rigor mortis (stiffness), and any obvious wounds or marks. When the M.E. arrives, they will do a more thorough examination and eventually transport the body for autopsy. Agents may assist by collecting trace evidence off the body at the scene (clippings of nails, combing hair for fibres, etc., especially if an FBI ERT is handling evidence collection). The agent will ensure any evidence on the body gets logged (for instance, if a knife is found still in the body, that will be collected at autopsy). The key is close coordination so that evidence on the victim isn't lost during the handling and transport.

By following these procedures – secure, document, collect methodically, and maintain chain of custody – FBI agents ensure that physical evidence retains its maximum value. It's this evidence that will later be analysed to provide investigative leads or courtroom proof. Next, we'll look at what happens to that evidence in various forensic disciplines and what an agent would know about each.

Fingerprint Evidence (Latent Prints)

Fingerprints are a cornerstone of identification in investigations. Every person (except identical twins to some extent) has unique fingerprints, and these patterns do not change over a lifetime. By the 1990s, the FBI have more than 200 million fingerprint cards on file – a massive repository started in 1924 – and fingerprints had long been used to match suspects to crime scenes. FBI agents in the field are well-versed in both the collection of fingerprints at crime scenes and the process of getting them identified.

- **Terminology:** A *latent print* refers to a fingerprint left by deposits of oils and sweat, usually invisible until developed. A *patent print* is one that is visible without enhancement (for example, a greasy fingerprint or a bloody handprint on a wall). A *plastic print* is an impression in a soft material like wax or wet paint. Typically when agents talk of "lifting prints," they mean latent prints.

- **Development Methods:** At a scene, as mentioned, common methods to reveal latent prints include dusting with fingerprint powder and using clear tape to lift them. Another common field technique, especially for non-porous surfaces, is superglue fuming (cyanoacrylate fuming). Agents might have a portable fuming kit – essentially a chamber or even a large trash bag: they put a small metal dish with a few drops of cyanoacrylate glue, heat it with a portable heater or just hot water, and let the fumes fill the enclosure with the evidence item. The fumes stick to fingerprint residue and polymerise, making a white print. After fuming, the print can often be enhanced with powder or fluorescent dye. For porous surfaces (paper, cardboard), chemicals like ninhydrin are used (usually back at the lab) – ninhydrin spray turns fingerprints purple by reacting with amino acids. Another chemical, silver nitrate, reacts with salts to make prints visible (an older method, less used by the '90s except on paper where other methods fail). Iodine fuming is yet another old-school technique: iodine crystals sublimate to vapour that temporarily reveal prints as brown (but they fade quickly, so you must photograph them fast). Agents probably recall these from training but in the field they'd more often collect the item for lab processing unless immediate results are needed.

An FBI ERT agent dusts a door for latent fingerprints while colleagues look on. In our campaign's era, fingerprint lifting is meticulous work – dust adheres to sweat residues and reveals the telltale whorls and ridges. Field agents might carry a small fingerprint kit (brush, powders black or gray, tape, lifter cards) for quick checks. But often, they'll summon specialised ERT technicians or let the lab handle complex surfaces. Fingerprints can be obtained not just from obvious places like door knobs, but off bodies (e.g. a bruise in the shape of a hand might be processed with special techniques), off tape or plastic bags, and even on difficult surfaces like the sticky side of duct tape (which requires chemical processing in a lab with dye stains).

- **AFIS and Identification:** Once a good print is lifted and photographed, the goal is to identify whose finger made it. Up to the 1990s, this was primarily done by expert examiners comparing the latent print to known prints on file. The FBI's Integrated Automated Fingerprint Identification System (IAFIS) was in development in the '90s and would go live in 1999, finally computerising the matching of prints. In 1997, however, the process is partly manual and partly automated on a smaller scale. Many state and local police have their own AFIS computers by the early '90s where a latent print can be scanned and searched against state databases of known criminals. The FBI's national database is still largely a warehouse of paper cards, though by 1995 the FBI had begun scanning those prints in preparation for IAFIS. Agents in 1997 will typically send latent print evidence to the FBI Lab's Latent Print section, where examiners will attempt to match it. If they had a specific suspect, they will compare to that person's fingerprint card. If not, they can request a search – the FBI can search the print against a growing digital repository or circulate it to likely jurisdictions. This process takes time; it is not the instantaneous computer hit you might see in the future of the 2020s. In the late '80s it took weeks or even months to get a fingerprint match from FBI records due to backlogs. By the late '90s the turnaround is improving as automation increase, but it can still be days to weeks. Agents will know to temper expectations and focus on other leads while waiting.

- **Expertise:** Agents are not certified latent print examiners (that's a specialised role). But with 5-10 years on the job, an agent knows how to read a fingerprint classification (the old Henry system for filing prints by pattern types), and they know the basics of what makes a match. Fingerprint examiners look for minutiae points – specific ridge characteristics like bifurcations (a ridge that splits) or ridge endings. Typically 8–16 matching minutiae might be required to call it a match (the exact standard varied). An agent reading the lab report in 1997 would see language like "Latent print Q-1 was identified as the right index finger of John Doe" or "No identification could be made". Agents also know that prints can be tricky – a crime scene might yield many partial or smudged prints. Not all will be usable. Additionally, it's common to find prints from innocent sources (family members, first responders, etc.), so context matters. They might have to collect elimination prints (fingerprints from known people who had legitimate reason to be at the scene, so the lab can exclude those).

- **Palms and more:** By the late 90s, law enforcement also recognise the value of palm prints and even bare footprints. The FBI fingerprint repository includes these if taken during arrest booking. A bloody handprint at a scene, for example, might actually be a palm, which can also be identified. Experienced agents know to lift anything that looks like ridge detail, not just fingertips.

- **Historical note:** As of 1997, fingerprints are still considered one of the most reliable forensic identifiers (DNA is relatively new by comparison). The FBI's fingerprint program is a point of pride; it has been running for over 70 years. A big change is on the horizon, though: in 1999, the new IAFIS will allow law enforcement anywhere to submit prints electronically and get a match in minutes rather than mailing cards and waiting weeks. But since our campaign is set in 1997, agents operate just before that leap – meaning some tasks (like driving down to the PD to use their state AFIS or mailing a fingerprint card to FBI HQ for a national search) are still part of the job.

In summary, fingerprint forensics in the late 90s involves a mix of traditional dust-and-lift groundwork and gradually improving database tech. Our FBI agents will be savvy about dusting a crime scene for prints and will know how critical it is to get good prints to the lab. They'll also be aware of fingerprint limitations – e.g. on some surfaces like cloth you just can't get a latent print, and not every crime yields useful prints (wearing gloves is common for perpetrators). Nonetheless, whenever we do find a clear print, that's a major lead.

DNA Analysis and Biological Evidence

By 1997, DNA forensic technology is revolutionising crime investigation – often called "genetic fingerprinting," it can uniquely identify individuals (except identical twins) from biological evidence. An FBI violent crime agent in 1997 will certainly have heard of high-profile DNA cases (like the O.J. Simpson trial in 1995) and likely has worked cases where DNA was tested. However, DNA analysis then is not as fast or routine as in 2025, and it requires sufficient sample and lab time. Here's what our agents know:

- **What DNA can come from:** Common sources of DNA at violent crime scenes include blood, semen (in sexual assault cases), saliva (cigarette butts, bite marks, envelopes), and hair roots. Essentially any tissue or fluid with cells can potentially yield DNA. Even dandruff or perspiration can have skin cells, though at the time "touch DNA" (getting DNA from just fingerprints or skin contact) is not yet common – it usually needs a visible biological stain. Hair without the root (just the shaft) has only mitochondrial DNA, which we'll get to shortly. Bone and teeth can yield DNA too, in cases of decomposed bodies (though that's more specialised).

- **Collection & Preservation:** Agents collect these as described: sterile swabs for wet stains, or cutting out dried stains. It's drilled into agents to avoid contamination – use clean gloves/tools, don't sneeze or cough on evidence, package items separately. In 1997, labs are able to get DNA profiles from surprisingly small samples (a few drops of blood or a tiny semen stain), but not as small as modern capabilities. Also, DNA degrades with heat, sunlight, and time. Agents in Phoenix must be particularly mindful that the desert heat and intense UV sunlight can break down DNA quickly. Bloodstains left in 110°F sun for days might become useless. So they'll prioritise collecting biological evidence promptly and storing it cool. Typically, after collection, bloodstained swabs or items are refrigerated or frozen until analysis to preserve the DNA.

- **DNA Profiling Methods (1997):** There are two main lab techniques in use:

  - <u>RFLP (Restriction Fragment Length Polymorphism)</u>: This was the first generation method used from late 1980s through the 90s. It requires a relatively large blood stain (like a dime-sized stain or bigger) and takes several weeks. RFLP looks at certain regions of DNA where patterns repeat and cuts them with enzymes to produce a band pattern on a gel – like a barcode unique to a person. By 1997, RFLP is still considered the gold standard for high-quality samples because it is very specific (examining multiple regions to virtually exclude anyone else).

    - PCR-based testing: PCR (Polymerase Chain Reaction) is a newer, faster method that amplifies tiny amounts of DNA. In the early 90s, a PCR test called DQ-alpha is used (it is a kit test that can yield results in a day or two, but it isn't as unique – it gives a genotype that many people might share). As the 90s progress, Short Tandem Repeat (STR) analysis is developed – this is the basis of modern DNA profiling, and it's PCR-based. By 1997, the FBI and other labs are validating STR markers (the FBI selects 13 core STR loci that will become standard for CODIS[1]). Some labs are starting to use STRs, but it is still early; widespread use will come around 1998-2000. An FBI agent in '97 might not know the term "STR" unless they keep up on lab developments; they'd just know the lab can do a DNA test with a faster turnaround if sample is small, likely referring to PCR tests. These tests can be done in a week or less in urgent cases, whereas RFLP might take 4–8 weeks including probe hybridisation and all.

- **Mitochondrial DNA (mtDNA):** Uniquely, the FBI Laboratory is a pioneer in mtDNA analysis in the mid-90s. Mitochondrial DNA comes from the mother's side and is found in cell mitochondria (as opposed to nuclear DNA). It's not as absolutely unique as nuclear DNA (because maternal relatives share it), but it can be obtained from hair shafts, old bones, or very degraded samples where nuclear DNA isn't recoverable. In 1996 the FBI brought mtDNA testing online for forensic use. So by 1997, if agents have a piece of evidence like a shed hair with no root, they might be aware that the FBI Lab can attempt mtDNA sequencing to see if it matches a suspect's maternal line. This will be more of a lab-to-agent communication; the agent just needs to know to send the hair in and request mtDNA if needed.

- **CODIS (DNA Database):** The concept of a national DNA database is just becoming reality. The FBI began a pilot DNA index system in 1990 with a few FBI state labs. In 1994, Congress authorised the FBI to create a Combined DNA Index System (CODIS) and national database of DNA from convicted offenders. By 1997, many states (including Arizona) are collecting DNA from convicted felons (often just sex offenders at first, then expanding) and using the CODIS software to store profiles. The national level of CODIS, called NDIS (National DNA Index System), is scheduled to go live in October 1998. So during our campaign year, the FBI is in the final stages of networking state databases. Practically, what does a Phoenix agent know? Likely: if they get an unknown DNA profile from a crime scene (say, a rapist's semen DNA profile), they can request a search in the DNA database. Arizona's state crime lab might run it against their offenders, and the FBI Lab could run it through the growing pilot national system. The agent should know it's not guaranteed – the database was much smaller then (mostly recent convicts), but it could hit a match to a previously convicted offender or link two cases together (DNA from Crime Scene A matches DNA from Scene B suggests the same perp).

- **Interpreting Results:** When the lab reports a DNA result, it often comes with a statistical weight. For example: "The probability of selecting an unrelated individual at random from the population with this same DNA profile is 1 in 6 million." In the '90s, RFLP provides very high random match probabilities (often in the millions or billions). PCR-based DQ-

1 Combined DNA Index System, see below.

alpha has lower discrimination (maybe 1 in a few thousand), but by combining multiple markers (like DQ-alpha with another marker Polymarker, or STRs), discrimination increases. An agent doesn't need to do the math, but they need to understand that a DNA "match" is not absolute certainty – it's reported with a likelihood. However, practically if you get a match with odds of millions-to-one, that's exceedingly strong evidence. If odds are lower (say 1 in 5,000), that's more a lead than proof by itself. Agents would likely discuss these with prosecutors and lab analysts to decide how to proceed.

- **Historical limitations:** In 1997, DNA tests can fail if the sample is too small or degraded. Also mixtures (like two people's blood mixed, or a sexual assault kit with both victim and perpetrator's DNA) are much harder to interpret than in 2025. Agents might get a result that says "mixture of at least two individuals; data insufficient for match". Additionally, turnaround time means DNA isn't a quick fix mid-investigation unless you expedite it. If you catch a suspect, you can get a known reference sample (blood from a suspect via consent or warrant) and send it to compare with scene evidence, but you might wait weeks for confirmation. So while DNA is a game-changer, agents still rely on traditional detective work concurrently.

- **Contamination concerns:** Agents in 1997 are being trained heavily to avoid contamination – by the 90s, lessons have been learned (some early DNA cases had issues where police accidentally mixed their own DNA). So an FBI agent will be careful: wearing gloves, using disposable instruments, and also they might collect elimination samples (for example, if a first responder bled at the scene, take their blood sample to exclude). They know that even sneezing on a sample could introduce foreign DNA. We keep mentioning it because it's really hammered into them: preserve the integrity of the biological evidence.

In role-play terms, your character would certainly consider DNA for relevant evidence. For instance, if you find blood that might be the perpetrator's (maybe the suspect was injured by the victim), you'd think "we need to collect that for DNA." If a victim was raped, you'd absolutely send a sexual assault kit to the lab for DNA testing. If you recover a weapon, you might look for blood traces for DNA or latent fingerprints. You'd also likely think to get reference samples – e.g., if you have a suspect later, you'll want their blood or a cheek swab to compare to the crime scene DNA. And if you have a victim and no suspects, you'd try the database route (CODIS) to see if the DNA matches any known offender.

One more concept: "chain of custody" for biological evidence means maintaining conditions too. Agents wouldn't leave a rape kit sitting in a hot car trunk all day – they know to get it refrigerated. In Phoenix, one might even bring a cooler with ice packs to a scene for temporary safe storage of collected DNA evidence until it can be moved to refrigeration.

So while DNA in 1997 isn't instantaneous, it's extremely powerful. Our FBI agents treat it almost like gold – something to be protected and utilised carefully. They also realise it's not in every case; many violent crimes have no perpetrator DNA left behind (e.g., a shooting from a distance). But when it is present, it can crack a case wide open, especially as databases grow.

Ballistics and Firearms Examination

When violent crimes involve firearms – and many do – FBI agents must turn to ballistics forensics, which is the examination of guns and ammunition-related evidence. In the late '90s, the science of matching bullets and casings to weapons is well established (dating back to the 1920s), but new computer databases are emerging to assist. Here's what our agents know and do regarding firearms evidence:

- **Firearm Evidence Basics:** If a gun is recovered, it will be test-fired by forensic examiners to compare its bullets and casings to those found at the crime scene. If bullets or shell casings are found but no gun, those items are still collected to glean information:

- **Rifling marks:** A bullet (projectile) can be examined for rifling marks. When a bullet is fired through a rifled barrel, the barrel's lands and grooves scratch a pattern of microscopic striations onto the bullet. These striations are like a fingerprint of the barrel – ideally unique. A trained firearms examiner uses a comparison microscope to look at the crime scene bullet next to a test-fired bullet from a suspect's gun. If the patterns line up in multiple consecutive striae, they declare a match, meaning that bullet was fired from that gun to the exclusion of all others. Agents will know in general that matching bullets is painstaking work – you need enough of the bullet intact (bullets often deform on impact) and a gun to compare it to.

- **Shells & cartridges:** A shell casing (from a pistol, rifle, or shotgun shell) can have several marks: the firing pin impression (a dent where the firing pin struck the primer), breech face marks (the pattern from the bolt face as the cartridge is slammed backwards), extractor and ejector marks (from semi-automatics yank and toss the shell), etc. These also can be matched to a specific firearm's mechanisms. If you only have casings and later seize a suspect's firearm, examiners can test-fire the gun and see if the casings it ejects bear the same microscopic marks as the evidence casings.

- If no firearm is recovered, bullets and casings can still yield clues. Calibre can be determined (from bullet diameter or cartridge dimensions). The firing pin shape can indicate what brand/type of weapon (e.g., Glock pistols leave rectangular firing pin impressions, distinctive from most other pistols' round ones). For shotgun shells, examiners look at the firing pin and breech marks too, and sometimes the wad[2] in a shotgun shell can be traced by type.

- Agents might also encounter tool marks if, for example, a bullet is removed from a wall with pliers or a suspect tampered with serial numbers on a gun. Firearms examiners also cover tool mark analysis in general, but we'll stick to guns here.

- **Gunshot Residue (GSR):** When a gun is fired, trace residues from the primer and gunpowder are deposited on the shooter's hands and clothes (and nearby surfaces). In 1997, a common method to test for GSR on a suspect's hands is to swab or use adhesive stubs on the person's hand soon after the shooting (the sooner the better, as it can be washed off or just fall off). The GSR particles (containing elements like barium, antimony, lead from primer) are then analysed in a lab, often with Scanning Electron Microscopy (SEM) coupled with X-ray analysis (EDX[3]), to confirm their presence and composition. Agents know that a positive GSR test on a suspect's hands can support that they recently fired a gun (or were very close to a gun fired). However, they also know the limitations: GSR can sometimes be inconclusive or absent even if someone fired a gun (perhaps they washed hands, wore gloves, or the ammunition had clean primers). Still, if dealing with, say, a subject who

2 plastic (historically sometimes fibre/cardboard) piece inside a shotgun shell that sits between the gunpowder and the pellets; when the shell is fired, the wad seals gas behind the shot and carries the pellets down the barrel. After exiting the barrel the wad usually peels away and falls to the ground, which is why it can often be collected as evidence at a shooting scene.

3 Energy Dispersive X-ray spectrometry

claims “I never fired a gun,” a GSR test might catch a lie. By the late 90s, the older paraffin test (from mid-20th century) is obsolete – that was a colour test that often gave false results; FBI agents will use the more reliable SEM-EDX method via the lab.

- **Trajectory and Scene Analysis:** Agents at the scene might attempt basic trajectory analysis if bullet holes are present. Stringing or lasers can be used: running string through bullet holes in walls or using rods in holes can show the bullet’s path, helping determine where the shooter was. In outdoor scenes, agents might look for impact points (a nick on a curb, a bullet embedded in a tree). If multiple shots, mapping trajectories might even indicate positions or order of shots. These tasks are often done by specialists, but an experienced agent can do preliminary work – e.g., using a laser pointer[4] placed in a bullet hole to see the bullet’s line. At minimum, they document where bullet holes are and recover bullets from backstops (with minimal damage, as mentioned).

- **Ballistics Databases:** A cutting-edge tool in 1997 is the use of computerised ballistics imaging to link cases:

    - The FBI had a system called DRUGFIRE, launched in 1993. It allows crime labs to capture images of cartridge casings and bullet markings and compare them on a computer. DRUGFIRE is especially good for casings. It helps link otherwise unrelated shootings by showing the same gun was used (for example, linking gang-related shootings across city lines). Phoenix, dealing with gang or drug-related violence, might very well use DRUGFIRE via either the FBI or local ATF office.

    - The ATF (Bureau of Alcohol, Tobacco and Firearms) had a similar system called IBIS (Integrated Ballistic Identification System) in the ’90s, which handled bullets particularly well. This separate development was an issue – two systems meant duplication. In 1997, the FBI and ATF agree to unify their ballistics databases, and this eventually leads to the National Integrated Ballistic Information Network (NIBIN) around 1999–2000. NIBIN combines the tech so that by the 2000s, all police can enter bullet and casing images into one national network.

- As FBI agents in 1997, you’d be aware that your lab or partners can run comparisons. For instance, if you recover shell casings from a murder scene, you can ask the FBI Lab or ATF to input them into DRUGFIRE/IBIS to see if those casings match any from other crime scenes. If a match is found, it’s a lead linking those cases (a big deal if it identifies a serial shooter or connects your case to another jurisdiction’s case). You might know the term NIBIN emerging, but practically, you’d just request a “DRUGFIRE comparison.” It’s worth noting the agent wouldn’t directly do this; they’d send the evidence to the lab which has the equipment.

- **What Agents Do vs Lab:** The field agent collects and documents, the lab does the microscopic matching. However, agents often confer with the firearms examiner. If a bullet is matched to a suspect’s gun, the examiner will tell the agent and provide a report stating the match (or non-match). Agents should understand that examiners also test functionality of guns (to see if it’s operable, or if it has a hair trigger, etc.) and measure trigger pull weight, etc., if relevant. For example, in a claim of an accidental discharge, an examiner might

4 The first red laser pointers became commercially available in the 1980s but were quite expensive then (often over $100). By the mid-1990s, prices have dropped significantly (many models under $50) and are used by, among others, FBI ERTs. Green laser pointers came a bit later, in the early 2000s.

comment on whether the gun has any defect that could cause that. Agents will include those findings in the investigative case file.

- **Serial Numbers and Gun Traces:** For firearms, another aspect is tracing ownership. Agents will record the serial number of any recovered gun and run a trace. The ATF handles firearm traces, contacting the manufacturer then distributor then dealer to find the original sale, and often finding if it was reported stolen or resold. FBI agents would coordinate with ATF or use their databases (or even the National Crime Information Center – NCIC – to check if gun is stolen). In the '90s, NCIC could be used to enter the serial number and see if the gun was listed as stolen or used in a crime.

- **Ballistic Evidence in Court:** Agents know that ballistic matches by examiners will be presented in court often with comparison microscope photos. In 1997 the general acceptance of bullet matching as evidence is high (the challenges to forensic science that came in the 2000s haven't yet cast doubt; juries typically trust examiners saying "it's a match"). FBI agents would have testified or seen colleagues testify about matches. They might use analogies like "it's like matching a fingerprint" when explaining to laypersons, though technically fingerprints are unique and ballistic marks, in rare cases, could coincide by chance or manufacturing similarities. But practically, agents in 1997 have confidence in their firearms lab colleagues.

In the field for our game, if there's a shooting, expect your agents to carefully recover all ballistic evidence. You'd make sure to collect shell casings (count them, so you know how many shots at minimum were fired), any bullets or fragments (perhaps found in walls or bodies – the latter will be retrieved at autopsy if not found at scene). You'd note damage from bullets (helps figure trajectory). If you find a gun, securing it safely is top priority, and later you'll want test-fires. You may also decide to do things like distance determination – if a victim has a gunshot wound, the presence of stipple (powder burns) or soot on the skin/clothing can indicate a close-range shot. The FBI lab's chemistry unit can test chemically for gunpowder residues on clothing to see if a shot was contact or near-contact (the classic "Greiss test" for nitrites, see GSR, above). An agent should be aware of this: if a dead victim has a firearm near them and it looks like suicide vs murder question, for example, you'd check for stippling or residue to confirm if it was a close self-inflicted shot or possibly staged. These are things the agent would mention and coordinate with the lab or medical examiner.

Summing up Ballistics: Our 1997 FBI agents trust that firearms evidence can link a suspect to a gun and a gun to a crime. They deal with it methodically: collect, document, send to lab. They also exploit new databases to find connections between crimes. In Phoenix, where firearms are common, they will likely have many cases involving ballistic evidence. They should also recall the importance of safety – always assume a gun is loaded until cleared, and be cautious at scenes (one never knows if a perpetrator left booby traps or a round in the chamber).

Trace Evidence: Hairs, Fibres, and Other Microscopic Clues

"Trace evidence" refers to small, often microscopic material that can transfer during violent crimes – a classic concept stemming from Locard's Exchange Principle (the idea that any contact leaves a trace). FBI field agents are taught to be on the lookout for these tiny clues and to collect them properly because they can often tell the story of an encounter. In the late '90s, the FBI Laboratory has experts in microscopy for comparing hairs and fibres, and in chemistry for analysing paint, glass, and soil. Here's what agents know:

- **Hair Analysis:** Hair is a frequent piece of evidence in violent crimes (from struggles, direct contact, etc.). Agents finding hairs (especially ones that look out of place, like a hair in a victim's hand or on their clothing that is a different colour than the victim's) will collect them. Under the microscope, a hair examiner can determine if a hair is human or animal, and if human, from what body area (head hair and pubic hair have different characteristics, for example). They can also compare a questioned hair to a known hair sample from a suspect or victim. In 1997, the process is microscopic comparison – the examiner mounts the unknown and known hair on slides and compares characteristics: colour, length, thickness, medulla (inner core) pattern, scale pattern on the cuticle, pigmentation granules, etc. They can say if a hair is "consistent with" coming from the suspect or not. Historically, FBI hair examiners often gave strong testimony that a hair matched a person, although by 2025's standards we know hair microscopy isn't unique like DNA. But in the '90s, it's considered a valuable technique. Agents would regard a hair match as a significant lead. They also know hairs can indicate things like if the root is attached and has tissue (great for DNA testing), or if the hair was forcibly removed (a ripped out hair may have a stretched root or skin attached vs a naturally shed hair with a club-shaped root).

In 1996, the FBI started using mitochondrial DNA testing on hairs. So if the microscope says "this hair is consistent with the suspect" but you want more certainty, the lab can attempt mtDNA sequencing on the hair. If the mtDNA matches the suspect, that strengthens the association; if it's different, the suspect can be ruled out. Agents in the field mainly need to know to collect hairs carefully (with tweezers, paper packets) and to collect comparison samples from suspects (around 50 full-length hairs from all parts of the scalp for example, to represent the variation). That's usually done via warrant or consent if needed.

- **Fibre Analysis:** Fibres from clothing, carpets, or other textiles often transfer during close contact. For instance, if a victim was wrapped in a blanket, fibres from that blanket may be on the victim's body. Or an attacker's clothing fibres might be caught on a broken window they climbed through. Fibres are collected (using tweezers, tape lifts, or vacuum filters depending on situation). The FBI Lab's Chemistry/Toxicology or Microscopy units can compare fibres side by side under a microscope for colour, diameter, cross-section shape, etc. They also use instruments like a microspectrophotometer to precisely compare colours (dyes) or Fourier-transform infrared spectroscopy (FTIR) to determine the chemical makeup of the fibre (e.g. cotton vs rayon vs nylon). If a fibre from a scene and a fibre from a suspect's sweater have the same microscopic and chemical properties, an examiner might say they could have a common origin. This is usually class evidence – many garments could share fibre types – but if it's a unique fibre (say a rare carpet type), it could be significant. Agents understand the value: fibres can place someone at a scene or show contact between victim and suspect. In 1997, a famous example they might recall is how the Atlanta child murders case in the early '80s was solved largely via fibre evidence (unique carpet and upholstery fibres linking victims to the suspect's home and car). So the FBI emphasises collecting fibres if present.

- **Paint and Glass:** In violent crimes, these often arise in hit-and-run cases (paint from a vehicle on a victim or scene) or break-ins (glass fragments on a suspect's clothing). The FBI Lab maintains databases for automotive paint, for example the PDQ (Paint Data Query) database in collaboration with foreign agencies, containing known car paint layer compositions by make/model/year. In 1997, if an agent finds a chip of paint at a crime scene that's from a car, they can send it to the lab; the lab can tell them "this paint chip is consistent with a General Motors model X from the early 90s" etc. That narrows the suspect vehicle list. Glass can be analysed for refractive index and elemental composition; if a suspect has broken glass in their shoe tread, lab scientists can compare it to broken window glass from the crime scene to see if they likely came from the same source.

Agents will collect these by carefully picking up glass shards (very small ones might be collected by tape lifting a suspect's clothing). Paint smears on an object could be collected by peeling or cutting that section if possible. They're aware to package in such a way that prevents loss (tiny fragments in a folded paper bindle inside an envelope, for instance).

- **Soil and Minerals:** Soil on a suspect's shoes or car could link them to a crime scene (especially in a desert or unique terrain). The FBI had a Mineralogy lab historically; by the 90s they could do comparisons of soil colour, content (pollen, minerals) etc. If a body was buried in a remote area, soil on the shovel of a suspect might be matched to that gravesite's soil. Phoenix's desert soil might have particular mineral profiles or cactus pollen that an examiner could find on clothing. Agents know this is pretty specialised – they'd only pursue it if it seems critical (like suspect denies being in a particular remote location but their truck has distinctive red sandy soil that matches that spot). Still, if the case calls for it, they'd collect soil (e.g. scraping dried mud from a shoe).

- **Other trace evidence:** There are many odd traces that could be relevant: gunshot residue we covered under ballistics; explosives residue if a bombing (not typical violent crime squad unless domestic terrorism, then FBI Bomb techs and lab handle it); latent footprints (dusty shoe prints, etc., which can be lifted with electrostatic dust lifters or gel lifts – an agent might use these or call in techs if footprint evidence is present). Also bite marks on a victim: while technically a pattern injury rather than trace, these would be photographed with scales and might involve a forensic odontologist comparing casts of a suspect's teeth to the bite (bite mark analysis was used in the 90s, albeit now regarded with caution). Agents would likely treat a bite mark as an important clue and preserve it (photograph, perhaps make a mould at autopsy). And blood pattern (which we discuss next) is another form of trace interpretation.

- **Case Linkage through Trace:** One thing agents consider: some trace evidence can tie cases together similar to how ballistics or MO[5] can. Say a series of burglaries have the same odd blue fibres left at scenes – that suggests the perpetrator wears a certain garment each time. Or a certain glittery makeup flake is found on multiple victims of a serial offender. The FBI lab's Trace Evidence Unit might notice if two submissions have the same peculiar substance and alert the agents. In 1997, this kind of cross-case analysis is more manual (requires examiners noticing coincidences), but ViCAP[6] or other systems might help if agents report those details. It's something a thorough agent keeps in mind.

Agents in the field primarily need to recall collection and preservation: use clean tools, avoid cross-contamination (don't put two different hairs in the same envelope), and label properly. They also must recognise what might be important: e.g., stopping to think "Could there be cat hair or unique fibres here that don't belong?" – If a suspect had a pet, for example, pet hair on a victim could be a clue. Or vice versa, victim's hair on suspect. This attentiveness to detail distinguishes a good CSI-

5 Modus Operandi, Latin for "method of operation". In plain English, an offender's MO is the set of behaviours / techniques they use to successfully commit the crime and get away with it – how they approach the victim, what weapon they bring, how they control the scene, how they escape, how they avoid leaving evidence, etc. MO can change over time as the offender learns what works and what doesn't.

6 See the section on Behavioural Analysis Support (Profiling), below.

minded agent from an average one. In gameplay, showing that you scan for these small things and bag them will make your character convincing.

Lastly, a quick note on laboratory reports for trace evidence: Typically, the FBI lab will report something like "One brown head hair recovered from victim's jacket was microscopically consistent with the head hair standard of Suspect X." Or "Fibre from victim's shirt matched fibres from suspect's car seat in colour, diameter, and polymer type." They usually won't say "it's a definite match" (except in rare cases of something extremely unique), but in 1997 they might sometimes overstate (the culture then was confident). Just know the difference between *individualising evidence* (DNA, fingerprints, bullets to a gun) and *class evidence* (fibres, glass – which can only show could be same source but not unique). An agent's knowledge level: they likely trust the lab's conclusions but also know that, for example, many people could have similar fibres on them if the fibre is common.

Bloodstain Pattern Analysis

When violence results in bleeding, the patterns those bloodstains make can speak volumes about what happened. By 1997, Bloodstain Pattern Analysis (BPA) is an established forensic technique, and FBI agents receive at least an introduction to it during training (often more detailed analysis is done by specialists, but many investigators have working knowledge). Here's what an agent would consider:

- **What is BPA?** It's the examination of the size, shape, distribution, and location of bloodstains to infer the events that caused them. Blood behaves according to physics; understanding that can tell you if a stain was from a blunt impact, a gunshot, a bleeding victim moving around, etc.

- **Types of Bloodstains:** Agents learn to categorise:

    - <u>Passive stains:</u> like drops formed purely by gravity (dripping blood), or pools. For example, a series of round drops leading away might indicate someone walking while bleeding (and drops fell off at intervals). The shape of passive drops (round vs elongated) can show if the person was moving (elongated means some motion or an angled surface).

    - <u>Transfer stains</u>: when a bloody object touches another surface (like a bloody handprint or a smear). A handprint in blood can identify a perpetrator or victim's movement. Wipes and swipes show motion – e.g., a swipe might show someone dragged a body or wiped their bloody hand on a wall.

    - <u>Projected or impact stains</u>: from blood that is propelled by force. This is where a lot of analysis comes in:

        - Low-velocity spatter: larger droplets (4mm or more) usually falling off a weapon or wound, often from things like blood dripping or minor force.

        - Medium-velocity spatter: smaller drops (1-4mm), associated with blunt force trauma or stabbings – each impact can fling blood off a weapon in an arc. Cast-off patterns (blood flung from a swinging weapon) appear as linear trails of drops; an agent noticing parallel lines of spots on a ceiling might deduce the assailant swung an object repeatedly (each swing cast blood off).

    - High-velocity spatter: very fine droplets (less than 1mm, a mist-like spray). This is typically from gunshots or explosions – the force is so great it aerosolises blood. A close-range gunshot to a victim's head, for instance, produces a fine mist of blood. Agents learn that such misting is a hallmark of gunshot spatter. Also, with gunshots, there can be back spatter (blood directed back toward the shooter from a close-range entry wound) and forward spatter (outward from exit wounds).

  - Arterial spray: if a major artery is cut, blood may spurt with each heartbeat, creating an arcing wave-like pattern on walls (with the pattern's spurts diminishing over distance as pressure drops).

  - Expirated blood: blood that's coughed or sneezed out (often a fine spray with air bubbles, maybe mixed with saliva). If you see blood with bubbles dried in it, that might indicate the victim was alive and bleeding from mouth or nose and exhaled blood.

- **Interpreting Scenes:** Agents on scene will observe bloodstains and form hypotheses. For example:

  - If a victim is found in a small room with blood spatter on the ceiling and walls in arcs, an agent might suspect a beating with a weapon (like a pipe or bat) that caused cast-off. If the pattern shows, say, arcs in two different directions, that might suggest multiple swings or even multiple assailants (depending on angle).

  - Blood trail leading away means the victim or suspect moved while bleeding. Agents will follow a blood trail if safe – it could lead to evidence (the suspect's escape route, discarded weapon, or where the suspect collapsed if also wounded).

  - Void patterns: These are areas within a spatter field that are clear, indicating something (or someone) was blocking blood spray at the time. For example, an outline of a body on the wall in clean space surrounded by blood means the body was there during a bloodletting event and then moved. Agents might note a void in spatter to infer positions of individuals or objects. If a suspect claims self-defence, but the blood pattern void suggests the suspect was standing in a particular spot when the victim was hit, it can refute or confirm stories.

  - Angle of impact: Investigators can determine the angle at which a blood drop hit a surface by the shape (elongated drops have a tail that points in direction of travel). By measuring the length and width, you can calculate the angle of impact (width/length gives sine of angle). Doing this for several drops and tracing back lines of convergence can pinpoint where the blood originated in 3D space – often used for gunshot scenes to find the area where the victim was when shot. In practice, in 1997 this might be done with strings taped to wall at angles, converging to a point of origin.

- **Expert Involvement:** While many agents have basic BPA training, the FBI does have specialists and often local departments or independent consultants with advanced expertise might be called for complex scenes. By the 90s, there were courses and texts (e.g., by Herbert MacDonell, one of the pioneers) that agents might have been exposed to. So, an agent might do a preliminary read but also secure the scene for a proper bloodstain analyst if needed. But on their own, they'd still use common-sense analysis: "The lack of blood on the suspect's clothing versus the copious spatter at the scene suggests they may have changed clothes or were standing out of the way." Or "The victim doesn't have trail marks beyond the kitchen, so likely died where found rather than moving around after injury."

- **Documentation:** Blood patterns are thoroughly photographed and sketched. If an agent thinks a particular pattern is important (say a cast-off arc on the ceiling), they might mark it and ensure multiple angles are captured. Sometimes strings or lasers used for trajectory are photographed in place to show the analysis.

- **Limitations:** Agents know BPA isn't an exact science like DNA. It helps reconstruct events but is interpretative. Still, in 1997 it's widely trusted in law enforcement. The agent might have to explain the conclusions carefully if testifying ("Based on bloodstain patterns, it appears the victim was struck at least three times while moving toward the door," etc.). They won't call themselves an expert unless they have that background, but they may use the info to guide interviews (e.g., if a suspect says he shot the victim from across the room but blood spatter indicates a close range shot, the agent sees the lie).

In gameplay, you can use blood pattern knowledge to decide your actions: noticing a blood trail might prompt you to follow it or coordinate a K9 track[7] along it. Recognising a cast-off pattern might tell you what kind of weapon to look for (e.g., a long object like a bat rather than a knife may create wider cast-off arcs). Or seeing fine mist on the suspect's clothing might indicate he was near a gunshot impact (and perhaps you then know to test that clothing for victim's blood microscopically). This forensic insight can give your team leads that pure witness testimony wouldn't.

Forensic Pathology and Autopsy Insights

Every FBI violent crime agent works closely with medical examiners (forensic pathologists) when deaths are involved. While agents are not doctors, with experience they become quite knowledgeable about cause of death indicators, time of death estimates, and injury characteristics. This helps them interpret autopsy reports and ask the right questions. Here's what our agents keep in mind:

- **Autopsy Process:** When a murder or suspicious death occurs, the Medical Examiner will perform an autopsy – a thorough examination of the body. Agents may attend the autopsy (especially FBI agents on cases of federal jurisdiction) or at least speak with the pathologist after. They do this to observe any evidence collection (like bullets or trace evidence being recovered from the body) and to get immediate impressions. Typically, the ME will dissect the body, document injuries, and collect samples (toxicology, etc.). Agents ensure things like the victim's hands were bagged at scene (to preserve evidence), and they might bring any information to the ME (e.g., they might tell the ME, "We suspect the victim was attacked around 11PM last night" or "Look for possible sexual assault signs" to aid the exam).

7 A K9 track (often written "K-9 track") is when trained police dogs are brought in to follow a scent trail from a crime scene. In Phoenix in 1997, this could mean calling local PD or Sheriff's Office K9, or state Department of Public Safety (DPS), not necessarily an FBI dog — FBI field offices don't typically keep general patrol/tracking dogs on hand. The track is time-sensitive, heat-sensitive, and terrain-sensitive (desert heat and porous ground kill scent fast), so if you're going to run a K9 track, you do it early.

- **Time of Death Estimation:** A common question – when did the victim die? By 1997 the science of time of death is well established but still an estimate. Agents know the big three early postmortem changes:

    - Rigor mortis: the stiffening of muscles after death. It generally starts ~2 hours after death, peaks around 12 hours (body fully stiff), then dissipates by around 24-36 hours (as decomposition loosens muscles). Warm environments speed it up a bit, cold slows it. If an agent finds a body that's fully stiff in a 1997 Phoenix summer, they might guess death was roughly half a day prior (but they'd defer to the ME for official estimate).

    - Livor mortis (lividity): the settling of blood in the body due to gravity, causing purplish discolouration on whichever side is down. Lividity starts within an hour or two and becomes fixed by ~8-12 hours. Agents at a scene might notice lividity patterns – for example, if lividity is fixed on the victim's back, but the body is found face down, that's a clue the body was moved some time after death. (Maybe killed lying on back, then later repositioned).

    - Algor mortis: the cooling of the body. Generally, a body loses about 1.5°F per hour in normal room conditions after initial plateau. But Phoenix outdoor heat can complicate this (a body might actually gain heat in extreme sun, or remain warm longer). Agents usually leave algor measurements to ME (taking liver or rectal temperature). It's one data point among others.

Beyond these, the ME might look at stomach contents digestion (to see last meal timing) and the state of decomposition if longer time has passed. Agents in Phoenix often contend with rapid decomposition in summer (heat speeds decay and insect activity) or mummification if very dry conditions. They know a rough rule: in hot weather a body can bloat within a day or two. A study in Arizona showed summer heat causes quick bloating in 2-7 days, but the arid climate can then lead to mummification, slowing complete decay – bodies outdoors might not skeletonise until several months have passed. In cooler winter, a body might look relatively fresh for days, but come summer it decomposes fast. Also, scavengers (coyotes, etc.) can scatter remains. These factors mean the ME's time since death estimates in the desert can have large ranges. Agents understand that and gather all clues: last known alive time, maggot activity, etc., to help pin it down.

- **Cause and Manner of Death:** The *cause of death* is the specific injury or disease that killed the person (e.g., "gunshot wound to the chest causing haemorrhage" or "strangulation"). The *manner of death* is a classification: homicide, suicide, accident, natural, or undetermined. Agents really care about both – if it's homicide, it's their case; if a suspicious death gets ruled suicide or accident, the criminal case may close (though if something seems off, agents might still investigate further).

The ME's autopsy will detail all injuries: entry/exit wounds of bullets, stab wound depths and trajectories, signs of struggle (defensive wounds like cuts on forearms or bruised knuckles of victim), any petechiae in eyes (tiny burst blood vessels in eyelids common in strangulation due to pressure), etc. Agents often ask the ME questions like: "Was the victim sexually assaulted?" – the ME would report if there's evidence like genital trauma or semen (they often do a rape kit in suspected cases, collecting swabs). Or "Could this wound be self-inflicted?" – e.g., MEs can tell a lot about gunshot wounds: contact wounds have soot in them, muzzle imprints; suicide shots often are contact at temple or mouth, etc., whereas a homicide might be from a distance or angle not possible by the victim. If a suspect claims the victim attacked them, the distribution of wounds can confirm or refute (e.g., multiple stab wounds in the back indicate an attack, not an accidental self-defence scenario).

Agents have a decent grasp: for example, they know stab wounds can be “sharp force injuries,” and they often ask MEs if a wound can give clues to the weapon (blade length, single or double-edged – double-edged knives make symmetrical wounds, single-edged have one blunt end in the skin). Blunt force trauma injuries (like contusions, lacerations on scalp) might indicate what kind of weapon (a heavy pipe vs a fist vs a bat – sometimes pattern injuries on skin can even show the shape of the object). The ME might find internal injuries like subdural haematoma in the brain with no external head wound – maybe strangulation or suffocation signs. Agents rely on these findings to piece together the how of the crime.

- **Toxicology:** A full autopsy includes sending blood, urine, and tissue for toxicology testing. Agents should consider if drugs or poison played a role. In a violent crime, sometimes the victim might have been drugged (e.g., sedatives given by a killer to subdue someone). Or the suspect might claim the victim was high and attacked them first (toxicology can confirm drug use by victim). MEs often take samples of liver or eye fluid if the body’s older. Standard panels test for common drugs, alcohol. In 1997, labs could detect a wide array of substances, though designer drugs or unusual poisons might need special tests. Agents will specifically request testing if they suspect something (like insulin injection murder or cyanide poisoning – which has telltale smell of bitter almonds at autopsy in some cases). The ME might consult with a forensic toxicologist for obscure poisons. Agents know enough not to rule out poisoning if the body seems “too clean” externally. If the autopsy finds, say, high cyanide or arsenic, the agent then pivots the investigation accordingly.

- **Entomology and Decomp Indicators:** If the body is decomposed or skeletonised, the ME may call in forensic entomologists or anthropologists:

  - Forensic entomology (covered in the next section) can help determine time of death by examining insect life stages on the body. Agents in Phoenix know that with the heat, insect activity can be intense and fast – maggots might consume a body quickly. An entomologist analysing maggots or pupae can often estimate how long since eggs were laid, thus a postmortem interval. The agent might ensure that fly larvae are collected at scene and sent to an expert.

  - Forensic anthropology comes in if mostly bones are left. They can identify the victim (if unknown) by skeletal features, and determine injuries to bones (like knife nicks on ribs, bullet holes in skulls, etc.). Phoenix being near desert expanses, it’s possible bodies are found after months or years in remote areas, just bones. FBI might then involve the Smithsonian or their own anthropology consultants. Agents themselves would step back and let the experts handle skeletal analysis, but they are aware of the basics: from bones one can tell sex (pelvis shape, skull traits), approximate age (tooth wear, bone fusing), stature, and possibly ancestry (skull measurements can suggest ethnicity, though with caution). They’d know to look for signs like healed fractures (which could be matched to medical records to ID a body) or unique dental work (dental records are a prime means of ID – see Odontology below).

- **Odontology:** For identification or bite mark analysis, forensic odontologists (dentists) are used. If a body is unrecognizable by face/fingerprints (common in decomposed or burned cases), teeth become critical. Agents ensure dental X-rays from autopsy are compared to antemortem dental records to ID the victim. FBI agents might coordinate obtaining the missing person's dental charts for the odontologist. Also, if there's a bite mark on a victim, an odontologist can cast it and compare to a suspect's teeth. In 1997 this is fairly accepted evidence (famously, in 1979, Ted Bundy was convicted in part by matching his teeth to a bite mark on a victim). Agents would treat a bite mark as significant: photograph with scale, maybe swab it for saliva/DNA, and get a forensic dentist involved. They should also consider that sometimes perpetrators are bitten by victims – so check the suspect for any recent bite injuries if caught soon after the crime.

- **ME's Report and Interaction:** After autopsy, the ME issues a report detailing everything. Agents will use that to inform charges and court evidence. But often, they already got a verbal rundown right after the autopsy – e.g., the ME might say "It looks like cause of death was two gunshots to the back, one pierced the heart. Death would have been within minutes. Also, I found defensive wounds on his arms, so he likely tried to shield himself." That helps the agent understand the scenario: the victim was shot from behind, possibly fleeing or unsuspecting. The ME might also give the bullets to the agent (after extracting them, they turn them over as evidence, which then goes to ballistics lab). Agents should verify the chain of custody – e.g., sign a form when receiving bullets or evidence from the ME.

- **Mortuary and Further:** After autopsy, the body is released for funeral, but sometimes additional examinations (like if later info requires exhumation) happen. Rare but possible in long investigations; an agent would coordinate if that ever had to happen, presenting new evidence to justify it.

In everyday terms, our FBI agents won't perform autopsies, but they should think like a detective-medical hybrid when seeing a body. At the scene: note things like lividity, rigor, injuries. Later: read the autopsy and glean investigative leads (like if stomach contents show last meal at a certain restaurant – send agents to check who they dined with; or if toxicology shows sedatives, look into who had access to those drugs). Also, be mindful of understanding wounds: if someone's claiming self-defence, and autopsy shows victim was shot in back of head execution-style, that's a contradiction your agent would note. Being well-versed in pathology gives the agents credibility and helps avoid relying on Hollywood myths (e.g., knowing that "immediate rigor" isn't a thing or that silenced guns still make some noise etc., those little details keep role-play grounded).

Forensic Entomology (Insect Evidence)

Unique to certain environments and lengthier postmortem intervals, forensic entomology is the use of insect activity on human remains to inform an investigation. In a hot climate like Arizona, insects can be the key to estimating time of death after the first day or two, and can also indicate if a body was moved or disturbed. By 1997, forensic entomology was gaining respect in law enforcement, though only a few specialists existed (the FBI itself had one agent, Special Agent Wayne Lord, who was an expert in entomology, often consulted on cases). Here's what our agents understand:

- **How insects help determine time of death:** When a person dies and their body is exposed, certain insects arrive in a fairly predictable sequence or wave. The most notable are blowflies (the metallic green/blue flies you see around garbage or dead animals). They are often the first to find a body – within minutes or hours of death if outdoors. They lay eggs, especially in body orifices or wounds (moist areas like eyes, mouth, open wounds). These eggs hatch into maggots (fly larvae) that feed on the tissue. The maggots grow through stages (instars) and then pupate (form a cocoon) and later emerge as adult flies. The timing of these life stages depends on species and temperature. In Arizona heat, development is faster than in more temperate climates, as warmth accelerates insect metabolism. A forensic entomologist can collect maggots and pupae from a body and either identify their stage or let them develop into adult flies to identify species, then, knowing the weather, can estimate when eggs were laid – hence when death likely occurred (since flies usually lay eggs shortly after death, barring things like body being indoors or wrapped for a period).

For example, if they find many fly pupae and only a few larvae, it suggests the maggots have already fed and started to pupate, meaning at least a week or more since death (depending on species and temperature). If only small first-stage maggots are present, maybe only a day or two. In Phoenix summers, a body can attract masses of maggots that can skeletonise a body in a matter of a week or two. In cooler winter, it might take longer, and different species (some flies are active only in warmer months). Agents would not calculate this themselves, but they'd know to secure the expertise.

- **Other insects:** After flies, come beetles (like dermestid beetles that eat dried flesh and larvae), then perhaps other arthropods. By analysing the types present – for instance, if only coffin flies (which come later and can get to more sealed bodies) are there – an entomologist may deduce how long or if the body was inaccessible initially. Ants and cockroaches can feed on bodies too; sometimes cockroaches nibble on skin and leave marks that might look like abrasions (an ME might mistake it for perimortem[8] injury without context). Agents might not know all the specifics, but they should remember to note insect presence (like "maggots present in wound, many flies around") and make sure samples are taken.

- **Indications of body movement:** Different insects prefer different environments. For example, certain flies might only be in urban settings, others in desert scrub. If a body has insects on it that are not native to where the body is found, it could mean the body was killed or stored elsewhere and then moved. A case example: if a body found in a dry desert area has insects that typically are found near water or indoors, that raises questions. One real case (not from Arizona): a corpse had cockroach activity that indicated it had been indoors then dumped outside. An entomologist helped solve the case by noting cockroaches wouldn't eat outdoors in freezing weather, meaning the body found outside was stored indoors for a while before being moved. An FBI agent may not deduce that on the spot, but they might later incorporate entomologist findings.

- **Cause of death clues:** Maggots tend to cluster at wounds or orifices because that's where they can easily enter. Maggots concentrated in an unusual spot (like on the chest or hands, not just mouth/eyes) indicates an unseen injury there. In one case, maggot placement tipped investigators off to stab wounds that were later confirmed on the skeleton. Agents hearing that story learn to pay attention: heavy insect activity around a particular area of the body might mean trauma there.

- **Collecting Insect Evidence:** Agents should coordinate to collect samples of insects at the scene. This might involve putting some maggots in a jar with alcohol (to preserve them for size analysis) and some alive (to allow them to mature for species ID). Also note the environmental conditions: temperature (they might even put a thermometer into the ground

8 refers to injuries or events occurring at or around the time of death. It comes from Latin: "peri-" (around) + "mortem" (death). In forensic investigations, this is distinguished from: *Antemortem* - injuries that occurred before death (while the person was still alive) and *Postmortem* - injuries or changes that occurred after death. The exact time frame of "perimortem" can be somewhat ambiguous and might span from shortly before death to shortly after.

or body at scene for entomologist reference) and whether the body was shaded or in sun. All that data is used by the insect experts. If the FBI's own entomologist isn't available, they might contact a local university entomologist – by the '90s, many states have someone who could help.

- **Scenarios in Phoenix area:** Perhaps more bodies found outdoors in remote areas compared to cooler climates. Our agents would know desert insects: blowflies will still come (they're nearly everywhere). In very dry conditions, sometimes remains mummify which can slow or alter insect activity – flies like some moisture. But often, body fluids will provide it initially. Also, desert scavengers (like coyotes, vultures) can complicate things by scattering body parts. Entomology might still be gleaned from maggots left in remaining flesh or bone marrow flies etc. If a body is buried, certain flies (coffin fly) can burrow and get to shallow graves, but deep graves might have delayed or different bugs. All those are expert-level, but an agent might consider burying attempts by suspect and still call in entomology if insects got there once partially uncovered.

- **One of few specialists:** In 1997, as mentioned, only a handful of people like SA Wayne Lord in the FBI or some academic guys like Dr. Lee Goff in Hawaii or Dr. Neal Haskell consult widely. An agent might have to telephone such an expert or ship samples to them. They know it's a niche field – "not the silver bullet in every case," but extremely useful when needed. For role-play, if your scenario has a decomposed body, your agent would realistically think: "We need entomology analysis to nail down time of death." You would ensure the maggots are collected (maybe mention calling in a specialist or if any agent has training). Use correct lingo such as "instar" or just "larval stages" to sound legit. And you'd be aware that "gross" as it is, those squirmy maggots are evidence, not just something to recoil from. Perhaps your character even bags a handful with gloved hands because it's evidence!

Also, you might glean clues: "The body's pretty far gone, but it looks like mostly bones with dried skin – likely a few months old. There are pupal casings in the soil – maybe it's been through a couple of generations of flies". That implies time passage. If the game master has entomology clues, following up on them is something an FBI agent would do.

Forensic Anthropology (Skeletal Remains) and Odontology

As touched on above, when remains are skeletal or severely decomposed, forensic anthropologists and odontologists (dentists) come into play. The FBI doesn't have an anthropologist in every field office in 1997, but they often collaborate with experts (the Smithsonian Institution has a long partnership with the FBI for unidentified remains). Here's what our agents know in these scenarios:

- **Forensic Anthropology:** An anthropologist can determine a lot from bones:

  - <u>Identification clues</u>: Sex can be determined (pelvis shape is the most reliable indicator – females have wider pelvic inlet, etc.; skull features also differ on average – men have more pronounced brow ridges, mastoid processes). Age at death can be estimated by looking at growth plates (in sub-adults), or degenerative changes in adults (wear on joints, changes in the pubic symphysis of the pelvis, etc.). Stature is estimated from long bone lengths using formulas. Ancestry (race) can sometimes be suggested by skull measurements and dental traits, though it's not 100%.

    - Trauma analysis: Anthropologists examine bones for signs of trauma. They can tell if a fracture happened perimortem because fresh bone breaks differently (with certain patterns and no healing) versus old healed fractures or postmortem damage (which might look different in colouration or break pattern because bone dried out). They might identify gunshot entry/exit on a skull (bevelling pattern in bone), knife marks on ribs or nicks indicating stabbing, or blunt force like depressed skull fractures.

    - Other: They can sometimes tell if remains have been disturbed by animals (tooth marks) or environment (erosion, burning). Burned bones can indicate attempt to dispose by fire.

Agents, at a scene of buried or skeletal remains, would secure the area and likely let an anthropologist or trained ERT do a careful excavation (to not miss small bones or evidence). The agent should note things like if the burial seems hurried (shallow grave), if body parts are missing (maybe scattered by animals). They'd assist by documenting the context (associated artifacts like clothing remnants, jewellery, etc., which could help ID the person).

Once bones are collected, if ID is unknown, agents help by providing missing persons data for comparison. For instance, if anthro says "deceased was a white female, approx 25-35 years old, 5'5" tall," the agent will comb missing persons records for someone fitting that in that area/time frame. They'll then get dental records or prior x-rays of those candidates to compare.

- **Odontology (Dental for ID):** Teeth are incredibly durable. If fingerprints are not available (say hands are gone or prints not in system) and DNA is pending or not feasible, dental records are the next best method in 1997. Most adults have dental records (from dentists) and those contain charts of fillings, tooth extractions, root canals, etc. The forensic odontologist will compare x-rays or charts of a known person to the teeth of the body. Agents are responsible for obtaining antemortem records of suspected matches. They may contact the family dentist of a missing person to get copies of charts or x-rays, then deliver them to the ME or forensic dentist.

If a positive match is made (for instance, a unique filling pattern matches perfectly), the agent then confirms the identity and can inform investigators/victims' families, etc. This is routine enough that an agent basically expects to chase dental records for unidentified remains cases.

- **Odontology (Bite Marks):** In some violent crimes (particularly assaults or homicides with sexual components), perpetrators bite victims. In 1997, forensic dentists often compare the pattern of a bite mark (which might be photographed or even the skin cut out and preserved) to a mould of a suspect's teeth. They'd look at the arrangement, gaps, size of teeth, etc. If a suspect is in custody, agents may get a court order to take an impression of their teeth (like a dental mould) and high-res photos of their bite. Then the expert overlays transparencies or computer images to see if they match the bruises/marks. Agents know that a match could be powerful evidence – for example, "these bite marks on the victim are consistent with the suspect's dentition." However, bite mark analysis, while accepted in the 90s, is now known to be less reliable. But our agents wouldn't have that 2020s scepticism; they'd treat it as legitimate. They'd just ensure to use a proper board-certified forensic odontologist for it.

If in-game a suspect is observed to have an odd tooth (like a missing canine) and the victim has a bite mark that also shows lack of that tooth in the pattern, an agent could make an on-the-spot inference, but formally they'd get the expert. They would also swab bite marks for saliva DNA – by

’97, it’s possible to get DNA from saliva in a bite. So ideally they do both: DNA to try for match, and pattern match by dentist.

- **Cultural Sensitivity:** One more aspect if dealing with remains, especially in Arizona, is the possibility of Native American reservation land or ancient remains (not our main focus, but FBI has jurisdiction over crimes on tribal lands too). Agents might come across bones that turn out archaeological rather than a recent homicide. The FBI would then involve proper authorities (e.g., archaeologists, tribal reps under NAGPRA law for old remains). But as a field agent, you first treat all found bones as potential evidence of crime until proven otherwise (like if obviously an ancient burial, etc.).

In summary, when the grisly work of retrieving bones or comparing dental records arises, our agents tackle it systematically: secure remains, call experts, facilitate IDs. They respect the expertise but need to know results to drive the investigation. Once an ID is made, the case often shifts from “Jane/John Doe” to a known victim, which is a huge step in investigation (now you can explore the victim’s life, suspects with motive, etc.).

One more forensic tidbit: sometimes anthropologists can assist in facial reconstruction for unidentified skulls. The FBI had a program (as mentioned in FBI Lab info, the VICTIMS project) to create facial reconstructions. Agents might try that if they hit dead ends. They’d either have a clay reconstruction done or a sketch artist approximate the face. They might release that to public to get tips on identity. Our agents, being thorough, will not overlook skeletal remains as unimportant just because the person’s long dead. They know even decades-old murders (cold cases) can be solved with these methods, and indeed the FBI in the 90s has units looking at unknown remains nationally.

Questioned Documents and Forensic Linguistics

Violent crime investigations occasionally involve documents – e.g., a ransom note in a kidnapping, a threatening letter, or even the handwritten logs/diaries of suspects or victims. FBI agents in 1997 will consider involving the Questioned Documents Unit (QDU) of the lab whenever authenticity or authorship of documents is in question.

- **Handwriting Analysis:** If you have a handwritten note (like a ransom note, or a letter from someone claiming to be a serial killer taunting the police), the FBI’s document examiners can compare it to handwriting samples from a suspect. Agents know to collect exemplars – handwriting samples. These can be of two types: requested (you ask the suspect to write certain phrases, often containing similar words or letter combos as the note, under controlled conditions) and non-requested (existing writings like journals, signatures on forms, etc.). The examiner will look at letter formations, slant, pressure, spacing, and unique quirks. In 1997, it’s quite possible for an examiner to opine “in my opinion, Suspect A wrote the note” or “the handwriting is consistent with Suspect A”. Agents treat that as useful evidence, albeit needing corroboration. They also caution that disguised handwriting or very brief notes might not be conclusive.

- **Typewriting and Printing:** Many documents could be typed (typewriters are waning by late 90s, but still out there, and dot-matrix or early computer printers are used). The FBI QDU maintains reference collections of typewriter fonts and can identify typewriter make/model by the font and any imperfections in the typing. If a typewriter was seized from a suspect, an examiner can sometimes match it to a typed note by its “fingerprints” (minute defects on the typeface or alignment). Similarly, early computer printers (dot matrix, daisy wheel) leave distinctive patterns. However, by ’97, laser printers and inkjets are making it harder to individualise beyond maybe tracing the brand of toner or subtle marks. Agents might not know all technical details, but they’ll certainly send in any printed notes; the lab might at least identify what kind of printer or if two notes came from the same machine.

- **Paper and Ink Analysis:** The lab can analyse the paper itself (watermarks, composition) and inks (using thin-layer chromatography or other chemical methods to see what ink formulation it is, potentially matching a specific pen brand or whether two writings were done in the same ink). For instance, if a kidnap note has two different black inks, one might suspect it was altered at a different time. Agents would know to handle documents carefully (touch edges, wear gloves to not add prints or smudge anything). They would also not fold or write on them. Ideally, put them in document protectors. And crucially: don’t allow original documents to be contaminated – if needing to photocopy, they’d do it carefully and avoid staple/paper clip holes by using plastic clips.

- **Indented Writing:** A cool technique known to agents: ESDA (Electrostatic Detection Apparatus) which finds indentations on paper. If someone wrote on a pad and removed the top page (like a ransom note), the impression might still be on the page below. The FBI could use ESDA to recover those indented writings (it’s like a charged plastic film that makes indented writing visible). Agents securing a writing pad or notebook at a suspect’s place will send it intact to the lab for that reason. They wouldn’t tear off pages. In 1997, ESDA is definitely in use. It could yield the text of notes that the culprit tried to keep no record of except they pressed hard enough to indent the next sheet.

- **Forged or Fake IDs/Documents:** While less glamorous in violent crime, sometimes suspects use fake IDs or alter documents (like insurance forms, etc.). The document unit can check for alterations (different inks, erasures). If a ransom note has a fake signature, they can see if it’s traced or printed.

- **Forensic Linguistics:** This is a newer field by the 90s but is gaining attention (the Unabomber case in 1996 famously had analysis of the manifesto for writing style clues). Agents might consult someone to analyse a threatening letter’s phrasing to glean profile hints (education level, background, regional dialect). In 1997 the FBI’s BAU (behavioural Analysis Unit) sometimes do this in colaboration with analysts – not a main lab function though. But an agent might note distinctive phrases or misspellings in letters, which could match how a suspect talks or writes informally.

In practice, if our scenario has any written evidence, your FBI agents would evidence-bag it and likely say “let’s get this to QDU for analysis.” If time-sensitive (like an active kidnap), they may do some quick observations themselves: e.g., is it handwritten or cut-and-paste letters? If handwritten, do we have any known writing of suspects to eyeball? Are there obvious peculiar spellings (like writing “enforsement” instead of “enforcement” might hint at an accent or low education or a ruse)? They might not be experts but they can draw hypotheses.

They also will dust documents for fingerprints before sending for writing analysis. The sequence matters: if using ninhydrin on paper, that might interfere with ink analysis, so coordination is needed. The FBI lab could do both: often ninhydrin to get prints, then still do ink analysis if needed (they’d take a tiny punch of paper for chemistry if needed, usually from a non-critical area). Agents ensure copies are made for working, while originals are preserved for court and lab.

Forensic Toxicology

Though typically more in the realm of medical examiners or specialised lab units, an FBI violent crime agent benefits from understanding toxicology – the study of poisons and drugs in the body – in the event that their case involves a less obvious means of violence (poisoning) or simply to interpret the role of substances.

- **Drug-Facilitated Crimes:** A scenario might be a victim who was incapacitated by drugs (e.g., a kidnapper dosing someone with sedatives or a serial rapist using a date-rape drug like Rohypnol which is an emerging issue in the late 90s). Agents should be aware of such tactics. In 1997, Rohypnol (flunitrazepam) is gaining notoriety, and the FBI would know to test a victim's blood/urine for it if suspected. GHB is another, somewhat less known drug in 1997. So an agent investigating a weird case where victim "just passed out" unexpectedly might consider testing for sedatives or other drugs.

- **Poisonings:** Not common, but certainly in FBI history – the case of the Tylenol cyanide poisonings (1982) or occasional homicidal poisonings (antifreeze, arsenic by a family member, etc.). If your investigation finds no external injuries but a sudden death, you might lean on tox results. Agents likely know classic symptoms: arsenic causes gastrointestinal issues, cyanide can cause pinkish skin and smell of almonds, carbon monoxide causes cherry-red lividity, etc. But they won't rely on memory alone; they wait for lab tests. If foul play suspected, they ensure thorough toxicology (blood, urine, organ tissues) and might even suggest specific tests if clues (like a bottle of strange chemicals found at suspect's house – test for those in victim).

- **Postmortem vs Antemortem:** Tox will show what was in victim's system. They also might test any beverages or food from the crime scene if poison is suspected. Agents must collect those (the half-drunk tea, the syringe by the body, etc.). They also separate samples to preserve (like hair can preserve a history of drug use over time, something known even in the '90s in some cases).

- **Common things:** In violent crime, common substances found might be alcohol or drugs that influenced the crime (like the suspect or victim being intoxicated). That can be relevant for motive or understanding events. Agents might ask the ME: "What was the blood alcohol level? Any cocaine or other drugs in system?" If a suspect claims self-defence because the intruder was high and crazy, tox can confirm if the intruder/victim was on PCP or such (PCP use causing erratic violent behaviour – known in law enforcement).

- **Chain of custody in toxicology:** Agents ensure samples are labelled and sealed (much like other evidence) because those results can go to court. E.g., showing a defendant was drunk or drugged might aggravate or explain their behaviour.

- **Inter-agency:** The FBI lab could do toxicology, but often the local ME's lab or a state lab would handle routine drug screens. FBI might get involved if it's something exotic or if a specific expertise needed (like chemical analysis of a rare toxin). In 1997, the science of detecting certain poisons (like ricin or radioactive substances) is less advanced, but traditional toxins like cyanide, arsenic, strychnine are detectable with proper protocols. The FBI also have the Chemical Toxicology lab that can analyse things like explosives residue or unknown substances.

To apply this: if your storyline had, say, multiple family members dying of "unknown illness," an FBI agent might suspect a poisoner (especially if it's a federal angle, maybe product tampering or crossing state lines). They'd advise toxicological tests and possibly bring in chemical analysts to identify poison in a suspect's possession. Or if a suspect is a doctor or chemist, more reason to consider poison.

Behavioural Analysis Support (Profiling)

While not a "forensic science" in the physical evidence sense, the FBI's approach to violent crime in the '90s famously includes profiling – the Behavioral Analysis Unit[9] (BAU) at Quantico helped investigate serial violent crimes by analysing behaviour patterns. As field agents in Phoenix Violent Crime, you have access to these services for complex cases (like serial murders, sexual homicides, bizarre or motiveless crimes, etc.).

- **Criminal Profiling:** Agents might send case details to the National Center for the Analysis of Violent Crime (NCAVC) at Quantico. Experienced profilers (like those popularised in Thomas Harris' novels or the actual agents John Douglas, Robert Ressler, etc.) would review the crime scene photos, autopsy, victimology, etc., and provide a behavioural profile of the UNSUB (unknown subject, widely used FBI jargon). They might suggest likely characteristics (e.g. "male, age 25-35, history of domestic abuse, job allowing travel") and suggestions for interview strategies or media release strategies.

- **ViCAP:** The Violent Criminal Apprehension Program is a database where details of unsolved homicides, abductions, sexual assaults, etc., are entered. Agents in 1997 would use ViCAP to see if their case might be linked to others (like similar MO or signature). For example, if you have an unsolved murder with a specific ritualistic element, you could query ViCAP to find if any other cases nationwide are similar. It isn't as instant as today's systems; often it required analysts reading and connecting dots. But by feeding your case in, you help the larger effort and might get a call from another state saying "hey, we have one just like that." This is particularly relevant to serial offenders who cross jurisdictions (which is why FBI started ViCAP in the 1980s).

- **Threat Assessment:** The FBI's National Center for the Analysis of Violent Crime also had specialists for things like extortion letters, bomb threats, etc. They could analyse a written threat for the type of offender (could tie into documents/linguistics). For instance, a kidnapping ransom note's phrasing and demands might hint at whether the person is a newbie or experienced, whether they are organised or disorganised – which influences how the agent negotiates or pursues.

- **Scenarios for Use:** If our campaign sees a pattern like multiple bodies in the desert with similar injuries, the agents would probably involve BAU. They'd compile a case packet (crime scene photos, summaries, forensic reports) and send it up. Meanwhile they continue standard investigation, but the BAU might return a profile that refines suspect pools (e.g., "likely an unmarried male who lives alone, may have a military background given the style

9 Renamed from the Behavioral Science Unit in 1994. This change occurred when the FBI created the Critical Incident Response Group (CIRG) and reorganized its behavioral analysis operations. The BAU became part of this new organizational structure at the FBI Academy in Quantico, Virginia. The unit's core mission of studying criminal behaviour, developing profiles, and providing investigative support remained the same, but the reorganization allowed for a more specialized approach to different types of cases.

of knots used to bind victims," etc.). The agents wouldn't treat this as gospel, but as another lead or perspective.

- **Caution:** Agents know profiling is not magic; it won't name a suspect, only guide. They combine it with the physical forensics and good old interviews and canvassing. But in the late 90s, the FBI is proud of its profiling history.

In our writing, you can have your agents throw around terms like "This crime scene suggests a disorganised offender" if it's chaotic, or "the choice of victim indicates maybe a personal cause". It adds flavour of FBI thinking. Just recall, these are educated guesses, not concrete like lab results.

Regional and Climate-Specific Considerations

Working in Phoenix, Arizona, offers some unique forensic challenges and opportunities that might not occur in cooler or wetter climates. The desert environment, high temperatures, and local geography all influence how evidence is preserved or degraded. As Phoenix-based FBI agents, you have a bit of specialised knowledge on these fronts:

- **Extreme Heat and Evidence Degradation:** Phoenix often sees summer temperatures soaring above 100°F (38°C). Such heat can accelerate decomposition initially (body tissues break down and bloat faster in the heat) but also dehydrate remains leading to mummification. A body left outdoors in the Sonoran Desert sun might bloat within a couple days, yet the aridity could dry out the corpse, slowing the later stages of decay. Mummified skin can persist for quite a long time (months or even years) without fully skeletonising. This differs from a humid climate where moist conditions promote continuous rapid decay to skeleton in a few weeks. Agents in Phoenix know a body can "age" weirdly – appearing quite desiccated and leathery on the outside but still have internal tissues relatively preserved. This affects time-of-death estimates – you'd rely on entomology and other clues, because the usual appearance/time formulae might not directly apply. Also, the intense sun and heat are brutal on DNA: ultraviolet radiation can degrade DNA in bloodstains or tissues within hours to days. So, a blood droplet on a rock in direct sun might yield no DNA profile if not collected quickly. Agents should shield biological evidence from sun as soon as possible (even using their own shadow or a makeshift cover) and get it refrigerated. Films and photographs also need care (remember, this is long before the advent of digital cameras); if you leave a bag of film in a car in Phoenix summer, you'll cook it – agents keep evidence in cooler, climate-controlled conditions.

- **Desert Terrain and Trace Evidence:** The ground in Arizona can range from hard-baked clay to loose sand to rocky gravel. Footwear impression evidence can be hit or miss. In firm dirt or dust, you might get a good shoe print outline. In shifting sand, prints are quickly lost to wind or collapse. Agents carry soil casting kits that include a fixative spray to harden a sandy print before casting. If a crime happens in a remote area, agents might call in Border Patrol trackers or Native American scouts who have skills in following tracks on hard ground. It's not "forensic" in the lab sense, but it's a practical investigative method in the Southwest. FBI agents themselves get some training in tracking, but often rely on those specialists. Nonetheless, they'll document any tire tracks or footprints with photographs and casts if possible. Tire tread patterns on dusty roads can be compared to suspect vehicles – one might find, for example, a distinct tread and patch on a tire that matches tracks at a dumpsite of a body.

- **Scavengers and Scattering:** In the Phoenix area deserts, animals like coyotes, javelinas, and vultures can scatter remains. If suspects dump a body in the wild, agents might find only partial remains. This means a wider search radius for evidence – bones could be dragged off, clothing torn and carried. Agents often coordinate a grid search with many personnel in daylight (and watch out for snakes!). It also means when remains are found, agents look for gnaw marks on bones (which tell anthropologists the damage was postmortem by animals, not knife marks by a killer). Being aware of scavenging prevents misinterpretation (e.g., missing fingers might be animals, not necessarily torture by killer).

- **Local Flora:** The desert plants can be both evidence and hindrance. Cacti spines could snag fabric from a suspect's clothing. Palo verde or mesquite pollen might stick to a suspect's vehicle. Forensic palynology (pollen analysis) is an obscure field, but if, say, a suspect claims "I never went out to the victim's burial site," and you find rare cactus pollen on their clothes or car that only grows at that site – that's a clue. More commonly, burrs or cactus needles carrying fibres can be found. Also, agents will note if vegetation is disturbed (broken branches or crushed plants could mark where someone walked or a body was dragged). One particular hazard: cholla cacti (jumping cholla) that stick to anything that brushes by. If a suspect has cholla segments stuck on their shoe or pants, it places them in the desert brush. That's an associative evidence a Phoenix agent would notice (and probably grimace, because they know the pain of those spines!). The agent might collect those plant pieces as evidence too.

- **Water and Flash Floods:** Phoenix is dry most of the time, but during monsoon season sudden downpours cause flash floods in washes. A body or evidence left in a normally dry arroyo could be whisked miles away by a flash flood. Agents are aware of the timing – if a victim went missing before a big storm and found far from expected, water may have moved them or their belongings. Conversely, a flood can also bury evidence under sediment. So timing searches before rains hit (or returning after to see what got uncovered) is tactical consideration.

- **Border and Transient Population:** Phoenix in the 90s sees some cases related to border crimes (human trafficking, drug smuggling violence) and has a transient/homeless population in urban areas. An FBI violent crime agent might need to identify victims or suspects who are undocumented or transient, which is harder – fingerprints might not be in databases. That's more an investigative challenge than forensic, but it means agents sometimes rely on forensic art (sketches or reconstructions) to ID John Does, or use things like isotope analysis (only burgeoning in late 90s) to see where someone grew up from their hair/tooth chemistry. The latter is not common in 1997, though.

- **Unusual Forensics:** Arizona has had some unique forensic situations – for example, a case of adipocere in arid climate (adipocere is a soap-like substance that can form from fat decomposition, usually in damp conditions, but there was a case in a cave in Arizona showing even arid, micro-environments can produce it). Not an everyday thing, but an experienced agent might have heard that bodies in deep shade or caves here might mummify or form adipocere if there's slight moisture. It reminds them to not assume "desert means totally dry."

- **Firearms and Ballistics Locally:** Arizona has high gun ownership; crime scenes may have multiple firearms or ammo present. FBI agents coordinate with ATF and local police often. If a case has multiple crime scenes across different counties, Phoenix FBI can more easily facilitate linking via DRUGFIRE as mentioned earlier. Also, many weapons used might be common ones (so less unique ballistic signatures if mass-produced barrels). They might encounter firearms from across the border (some full-auto or military surplus weapons). The FBI lab can handle those, but agents should note markings if a gun has been filed (illegal guns might have serials obliterated – FBI can do serial number restoration with acid etching because the metal's crystalline structure is changed where stamping occurred, sometimes recovering the number).

- **Cultural factors:** Phoenix is near Native American reservations and has a significant Mexican-American population. Agents must be mindful if dealing on reservation land (they have jurisdiction for federal crimes there, often working jointly with tribal police). If a crime scene is on tribal land, the FBI will involve the tribal authorities and may need to adapt evidence collection to any cultural sensitivities (for instance, some tribes have traditions about handling the dead – the FBI must do their job but also show respect and explain what's happening to families). Language or cultural differences might come into play with witness statements or suspects (not strictly forensic, but for context – an agent might get a translator for a witness who speaks only Spanish to correctly document their statement, or use bilingual ability for interviews).

- **Environmental Dangers for Investigators:** Heat stroke is a real risk working a scene at noon in July. Agents would schedule processing in early morning or dusk if possible, and wear hats, hydrate, etc. It's not evidence-related, but it might come up in role-play that you choose to process a remote scene at 5 AM to avoid midday heat, which is realistic. Also, watch for rattlesnakes at scenes – yes, an agent poking around rocks for a shell casing should be cautious. They teach situational awareness for that in local law enforcement, and the FBI agent would likely have that local knowledge or get local officers' help who know the area.

In summary, being in Phoenix means you're somewhat of a "desert forensic" specialist by necessity. You adjust evidence tactics to the climate – faster collection of perishables, knowledge of how climate affects decomposition and DNA, and leveraging the environment (like unique soil or plant evidence) when it helps. These are the kinds of exotic tidbits that an agent in Seattle or New York might never think of, but in Arizona, they can make or break a case. Our campaign's agents can pride themselves on that know-how.

Conclusion: Putting It All Together

After absorbing this exhaustive guide, you can see that an FBI field agent in the late 90s had to be a bit of a renaissance person in forensics – not performing lab tests, but knowing what's possible, what to look for, and how to talk to the experts. In a complex violent crime scene, our agents will:

- Secure the scene and ensure evidence isn't contaminated or lost (whether that means taping off an alley or shielding blood from the desert sun).

- Methodically document everything with photos, sketches, and notes so that months or years later the scene can be reconstructed accurately from records.

- Collect physical evidence with care: latent prints, DNA, bullets, fibres, and more, each in proper packaging, maintaining the precious chain of custody.

- Leverage the FBI Laboratory’s full capabilities – sending evidence to Quantico (or D.C. – Quantico is about 45 min to an hours drive from Washington D.C., so to an Arizona resident they’re practically the same place) for analysis in disciplines from DNA to tool marks to questioned documents. They know roughly how long things take and plan investigations accordingly.

- Interpret preliminary forensic findings to guide leads: a fingerprint match might give a suspect name, a DNA hit could tie cases together, a ballistics link can show the same gun in two crimes, a fibre match might put a suspect in contact with a victim.

- Combine physical evidence with behavioural insights – using profiling and databases like ViCAP to see the bigger picture of serial offenders, and using forensic results to either support or refute witness accounts and suspect stories.

- Always keep in mind the timeline: what was tech in 1997 and what wasn’t. Our agents won’t have rapid DNA machines or fancy touch DNA tests; they deal with fax machines and pagers to get lab reports, and they may wait days for an AFIS fingerprint search or weeks for an RFLP DNA result. They have to make decisions with partial info often, and use old-fashioned legwork to supplement forensic waits.

Finally, historically accurate role-play means you might occasionally limit yourself: e.g., no calling in a cell-phone geolocation (not really a thing then beyond crude tower data), no checking someone’s DNA in an hour on a handheld device, etc. But you do have the advantage of the burgeoning FBI databases (fingerprints are about to go digital, DNA database in infancy, ballistics network being built). You’re in an era where forensics is on the rise, and as agents you’re excited to use these tools to solve crimes that might have stumped previous generations.

With this sourcebook digested, you should feel confident to handle any gritty crime scene your game master throws at you with realism and flair. Whether it’s lifting a fingerprint off a dusty 1970 Pontiac at noon on an August day, or coordinating an autopsy for a body found in a saguaro-studded canyon, you now have the know-how to proceed as a true FBI agent of the 90s. Good luck, and stay cool out there in the desert heat – both literally and figuratively!